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DNP3 In-Network Timing Normalizer — Tutorial

Concealing the Cross-Layer Response Time fingerprint on an Intel Tofino-1 switch

Source commit: 4d0df81 · Built: 2026-07-25 19:05 UTC · Reference program:
dnp3_timing_normalizer.p4 (sha 82f572ce)

1. What this tutorial is

In plain terms. A device on a SCADA network can be identified just by *watching* how long it takes to answer — no need to touch it. This project makes that timing look identical no matter which device is behind the switch, using the switch itself. This tutorial explains the idea from first principles, shows the mechanism on real hardware, and is honest about what it does and does not hide.

The tutorial has two layers. **Layer 1** (plain language) opens every major section — read just those if you want the idea. **Layer 2** (technical detail) follows for readers who want the P4 and the measurements. Every acronym is defined the first time it appears.

DNP3 = Distributed Network Protocol 3, a common protocol between electric-grid control rooms and field devices. *CLRT* = Cross-Layer Response Time, defined in §3. *Tofino-1* = the programmable switch chip this runs on.

2. Background: DNP3, masters, outstations, and the three messages

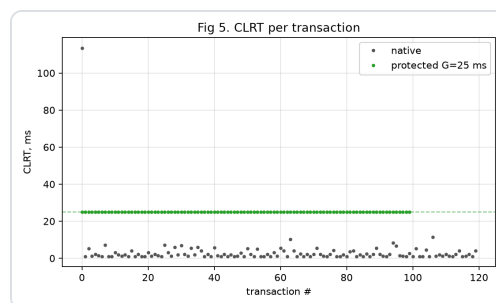
Layer 1. In DNP3 a **master** (in the control room) asks questions and an **outstation** (a field device such as a protective relay) answers. To read the outstation's data the master sends a **READ**. Because DNP3 runs over TCP, the outstation's computer first sends a bare **ACK** ("I received your packet") and then, a little later, the actual **RESPONSE** (the data). So one transaction is three messages: READ → ACK → RESPONSE.

Layer 2. The READ is a DNP3 application request with function code 1. The ACK is a pure TCP acknowledgement — a segment with the ACK flag set and zero payload (`tcp.len == 0 && tcp.flags.ack == 1`). The RESPONSE is a DNP3 application response, function code 129. On the relay we study (a Schweitzer SEL-751), the ACK and RESPONSE are **separate** packets; some other devices piggyback the ACK onto the response and so have no separate ACK at all — a distinction that matters in §11.

3. What CLRT is and why it fingerprints a device

Layer 1. The gap between the ACK and the RESPONSE is the time the device spent thinking — reading its inputs, building the answer. Different devices think at different, characteristic speeds, so this gap is like a fingerprint. Formby and colleagues named it the **Cross-Layer Response Time (CLRT)**.

Layer 2. The ACK is emitted by the TCP stack on receipt; the RESPONSE is emitted after the firmware's scan-and-assemble path. The gap is therefore a direct measurement of internal processing latency, stable per device and distinct across devices. On our physical SEL-751 the native CLRT has a median of 2.03 ms and a standard deviation of 10.33 ms; measured as observer entropy at 1 ms resolution it carries **2.73 bits** of identifying information.



Native CLRT varies transaction to transaction — the spread is the fingerprint.
Data: evidence/native/native120.pcap;
script: scripts/make_pub_figures.py.

4. The threat model

Layer 1. The attacker only *watches* the network — they never log into any device. From timing and packet headers alone they work out which relay model is answering, then look up that model's known weaknesses. The defender wants the wire to look the same regardless of the device behind it.

Layer 2. This is passive device fingerprinting: reconnaissance with no active probing. It is one step in a kill chain — model identification precedes targeted exploitation. The defense must be *non-cooperative* (it cannot modify the certified relay) and must not itself add a detectable software-latency signature.

5. The idea: hold the response inside the switch

Layer 1. When the real RESPONSE reaches the switch, the switch does not send it on immediately. It parks the *original* packet in a waiting area and releases it at a fixed time — the ACK time plus a chosen constant **G**. Every response then leaves at the same delay, so the timing fingerprint disappears. Crucially the switch never rewrites the packet: what comes out is byte-for-byte what the device sent.

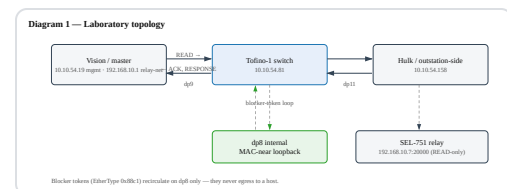


Diagram 1 — the lab: master (Vision), Tofino-1 switch, outstation-side host (Hulk), and the physical SEL-751 relay. dp8 is an internal loopback used only inside the switch.

Layer 2. The switch classifies the DNP3 transaction in its ingress pipeline, steers the response into a low-priority Traffic-Manager queue, and declines to schedule that queue until a data-plane deadline. No endpoint is modified; no controller is consulted per transaction (the control plane only sets **G** once). Because the original packet is enqueued unmodified and released as-is, the mechanism is byte-preserving — no CRC recompute, no field edits.

6. Where the original response waits, and what blocker tokens are

Layer 1. The waiting area is a queue inside the switch. To keep that queue “closed”, the switch keeps a small pool of tiny internal packets — **blocker tokens** — busy in a *higher-priority* queue. As long as those tokens are circulating, the switch always serves them first and the response queue has to wait. The tokens never leave the switch, so anyone watching the network sees none of them.

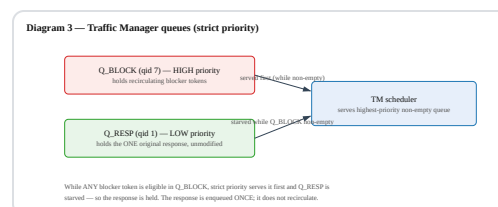


Diagram 3 — two queues under strict priority. While Q_BLOCK (high) has tokens, Q_RESP (low, holding the response) is starved.

Layer 2. *TM* = Traffic Manager, the switch’s queueing block. Q_BLOCK is queue id 7 (high strict priority); Q_RESP is queue id 1 (low). Blocker tokens are internal frames with EtherType `0x88c1` that recirculate on the loopback port dp8. A *reservoir* of tokens ($K \geq 64$) is required, not a single token, because a lone token can drain between arbitration cycles and briefly open the gate. “Holding” is therefore an emergent property of keeping the high-priority queue non-empty, not a software timer.

7. Why blocker tokens never reach the endpoints

Layer 1. The blocker tokens exist only to occupy the switch's scheduler. They loop around an internal port and are thrown away when their job is done. They are never addressed to the master or the outstation and never go out a real port.

Layer 2. `to_block()` sets `bypass_egress = 1` and directs the token to the loopback port, so it never traverses egress to a host. We verified this: the protected capture contains **zero** frames with EtherType `0x88c1` at the master. Confirm it yourself:

[Show the check](#)

8. How the deadline arms and releases the response

Layer 1. When the switch sees the ACK, it notes the time and sets a deadline: ACK time + G. While the deadline is in the future, the blocker tokens keep circulating and the response waits. The moment the deadline passes, the tokens are retired; the high-priority queue empties; the switch finally sends the response — untouched.

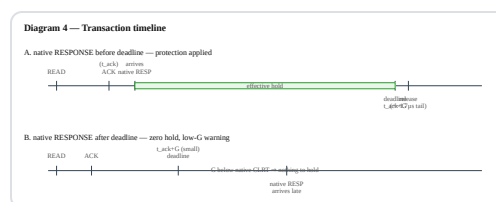


Diagram 4 — the timeline. Case A: the native response arrives before the deadline, so it is held to G. Case B: G is set below the native CLRT, so there is nothing to hold (a low-G warning fires).

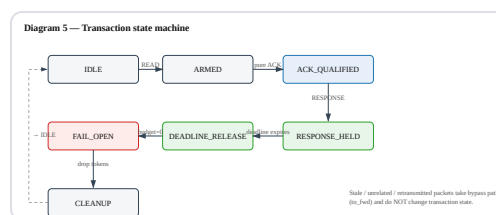


Diagram 5 — the per-transaction state machine. Unrelated or retransmitted packets take bypass paths and do not change state.

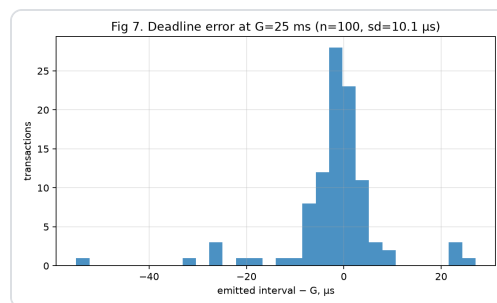
Layer 2. The transaction is armed by the READ (which claims a per-transaction *generation* tag). The first qualifying ACK writes the deadline **idempotently** — a retransmitted ACK cannot move it. Each recirculating blocker compares the current time to the deadline using a sign-bit ternary match (the chip cannot do a 32-bit magnitude compare in one step); past the deadline the token is

terminated instead of re-injected. Full arithmetic
and the packed state word are in

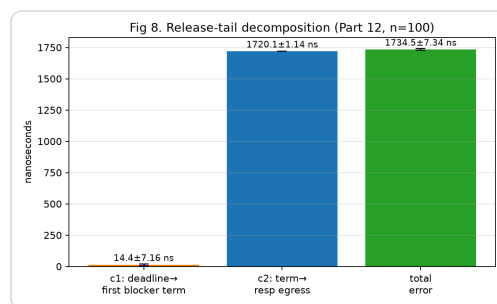
`CODE_WALKTHROUGH.md` §5–§10.

9. Why the release has a small, stable tail

Layer 1. Release is not perfectly instant — there is a tiny, *fixed* lag between the deadline and the response actually leaving. Because it is fixed (about 1.7 microseconds) and the same for every device, it carries no fingerprint.



Deadline error at G = 25 ms: the emitted interval clusters tightly around G. Data: evidence/protected/final100_g25.pcap; script: scripts/make_pub_figures.py.



Release-tail decomposition (Part 12, n=100): deadline → first-blocker-sees-it (c1 ≈ 14.4 ns) and termination → egress (c2 ≈ 1720.1 ns), total ≈ 1734.5 ns.

Layer 2. The tail decomposes into c1 (mean 14.4 ns, sd 7.16 ns) and c2 (mean 1720.1 ns, sd 1.14 ns), total 1734.5 ns with sd 7.34 ns — a property of the recirculation loop, independent of the device. It is why the protected CLRT sits at G + ~1.7 μs and why residual timing entropy is 0 bits at millisecond resolution, appearing only below 100 μs.

10. Fail-open and how G is selected

Layer 1. If anything goes wrong with the holding loop, the switch **releases** the response rather than trapping it forever — losing a real SCADA response is unacceptable. And G must be chosen *above* the device's natural response time; otherwise there is nothing to hold.

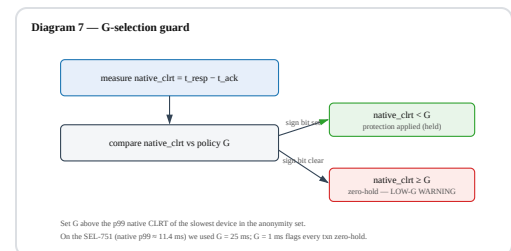


Diagram 7 — the G-selection guard: it measures the native CLRT and flags any transaction where G is too small (zero hold).

Layer 2. Fail-open: each token has a finite pass budget; a token that exhausts it retires the transaction and the response is released (`ctr_release_fail_open`). **G selection:** the guard measures `native_clrt = t_response - t_ack`, flags `protection = native_clrt < G`, and counts `zero_hold` transactions. Set G above the p99 native CLRT of the slowest device in the anonymity set — we used 25 ms (native p99 ≈ 11.42 ms). At G = 1 ms every transaction is flagged zero-hold.

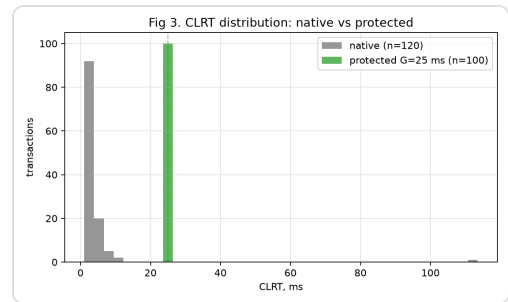
11. What the mechanism does and does not conceal

Layer 1. It hides the *timing* fingerprint — and only that. It does not hide the fact that this device sends a separate ACK, nor its TCP settings, nor its response size. So it is one piece of a bigger picture, not a full disguise.

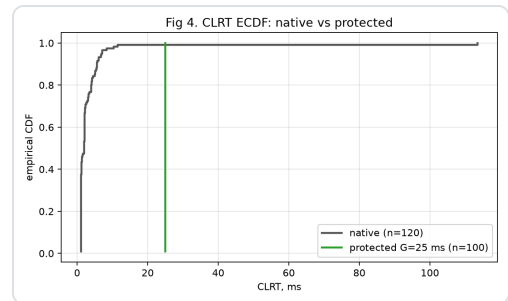
Layer 2 (claim discipline).

- **Conceals:** the CLRT-magnitude channel — entropy drops from 2.73 bits to 0.00 bits at millisecond resolution on the real SEL-751 traffic, with the response byte-identical.
- **Does not conceal:** *ACK mode* (the SEL-751 uniquely sends a separate ACK), *TCP-stack* signature (TTL/MSS/window), and *response size*. On our 3-device corpus ACK mode and TCP stack already identify the relay at balanced accuracy 1.000, and they are untouched here.
- Because only the SEL-751 has a CLRT at all in this corpus, closing it is an *anonymity-set-of-one* result — a genuine within-channel reduction and a working mechanism, but not, by itself, a reduction in a real multi-channel device classifier. Size obfuscation is a **separate, unproven** line, out of scope this week; this mechanism does not pad or split.

12. Results

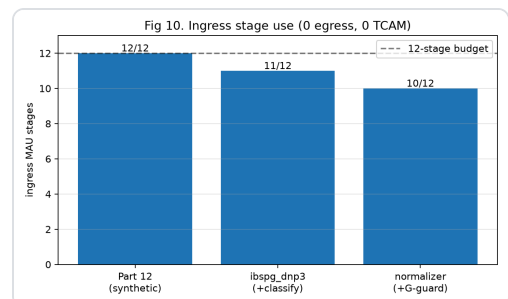


Native vs protected CLRT distribution. Native (grey) spreads from ~2 ms out past 11 ms; protected (green, G = 25 ms) collapses to a single spike. Script: `scripts/make_pub_figures.py`.



The same, as cumulative distributions.

At G = 25 ms over 100 repetitions on the relay's real frames: median 24.999 ms, standard deviation 0.010 ms (native 10.33 ms) — a 1000× spread collapse — and byte-identity on all 100 responses. The switch footprint is 10 of 12 ingress stages, 0 egress stages, 0 TCAM.



Ingress stage use across the program's evolution; the mechanism fits comfortably in the 12-stage budget.

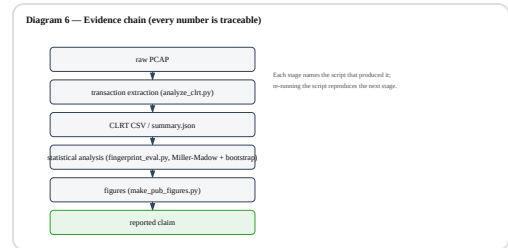


Diagram 6 — the evidence chain: every reported number traces from a raw PCAP through named scripts to the claim.

13. How to run it

The runnable lab lives in

`research/timing_final/` with a `Makefile`.

The safe default is a replay demonstration of the relay's real frames — no relay modification, no DNP3 control or write traffic.

The ten-minute path (see [QUICKSTART.md](#) for expected output per step)

The one guarded command is `make demo`

`MODE=replay TRIALS=10 G_MS=25`. Full operator procedure: `LAB_RUNBOOK.md`. Inspect the captures: `WIRESHARK_GUIDE.md`. If anything misbehaves: `TROUBLESHOOTING.md`.

14. Limitations and security scope

This is a **mechanism** result on real Tofino-1 silicon: a data-plane-scheduled, chaff-free, byte-preserving timing-normalization state that substantially reduces the CLRT-magnitude fingerprint of one physical relay. It is **not** full device anonymity, **not** size obfuscation, **not** a live inline relay session (it is replay of the relay's real frames), and it is not demonstrated across devices or TCP configurations, nor is it production-ready. Turning the within-channel result into an end-to-end security result requires a fleet of separate-ACK devices normalized to a shared G with ACK-mode and TCP-stack held constant — the natural next step. See

[TIMING_MECHANISM_EXPLAINED.md](#) and [TIMING_FINGERPRINTING_ANALYSIS.md](#) for the full treatment, and [references.md](#) for sources.